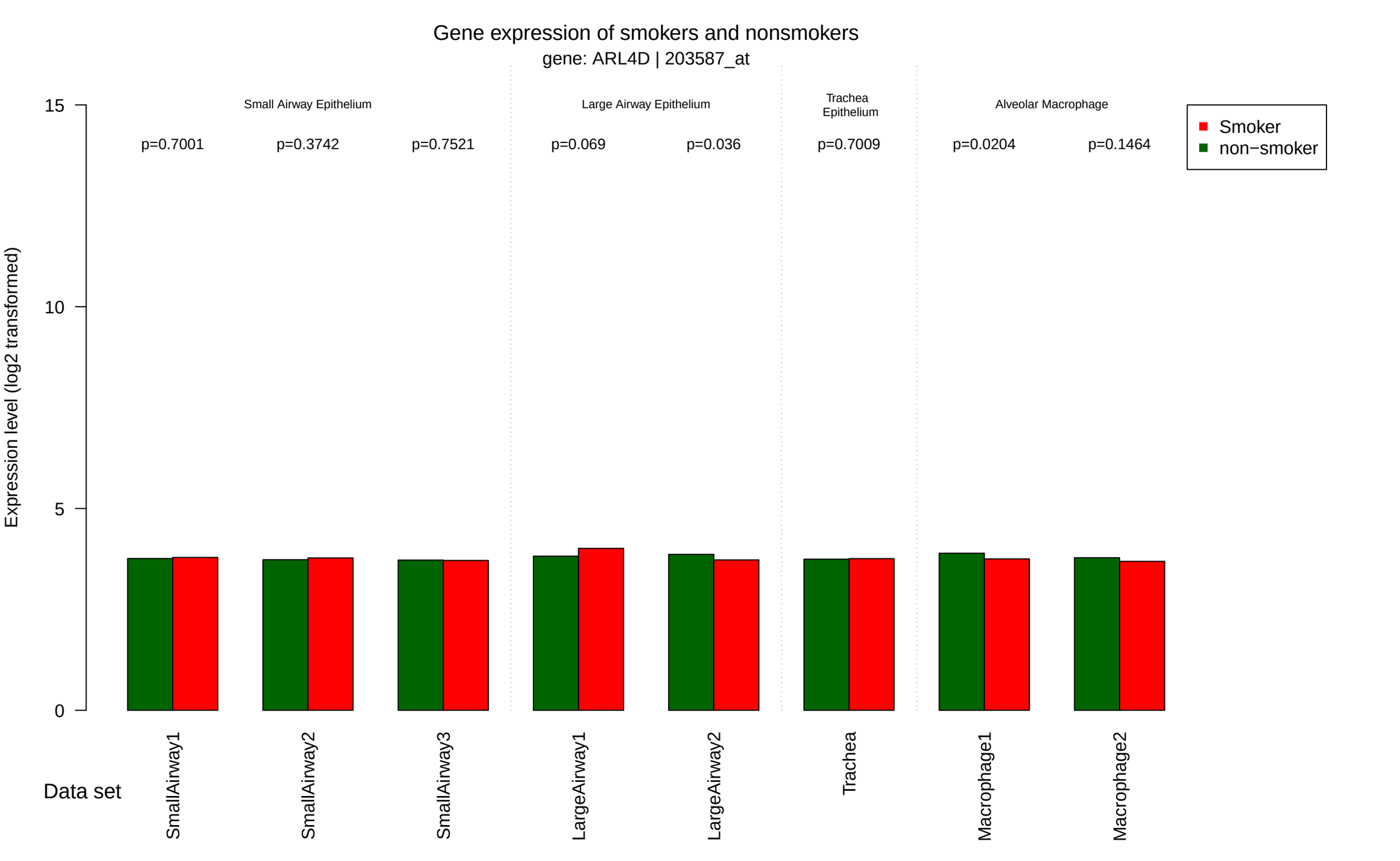
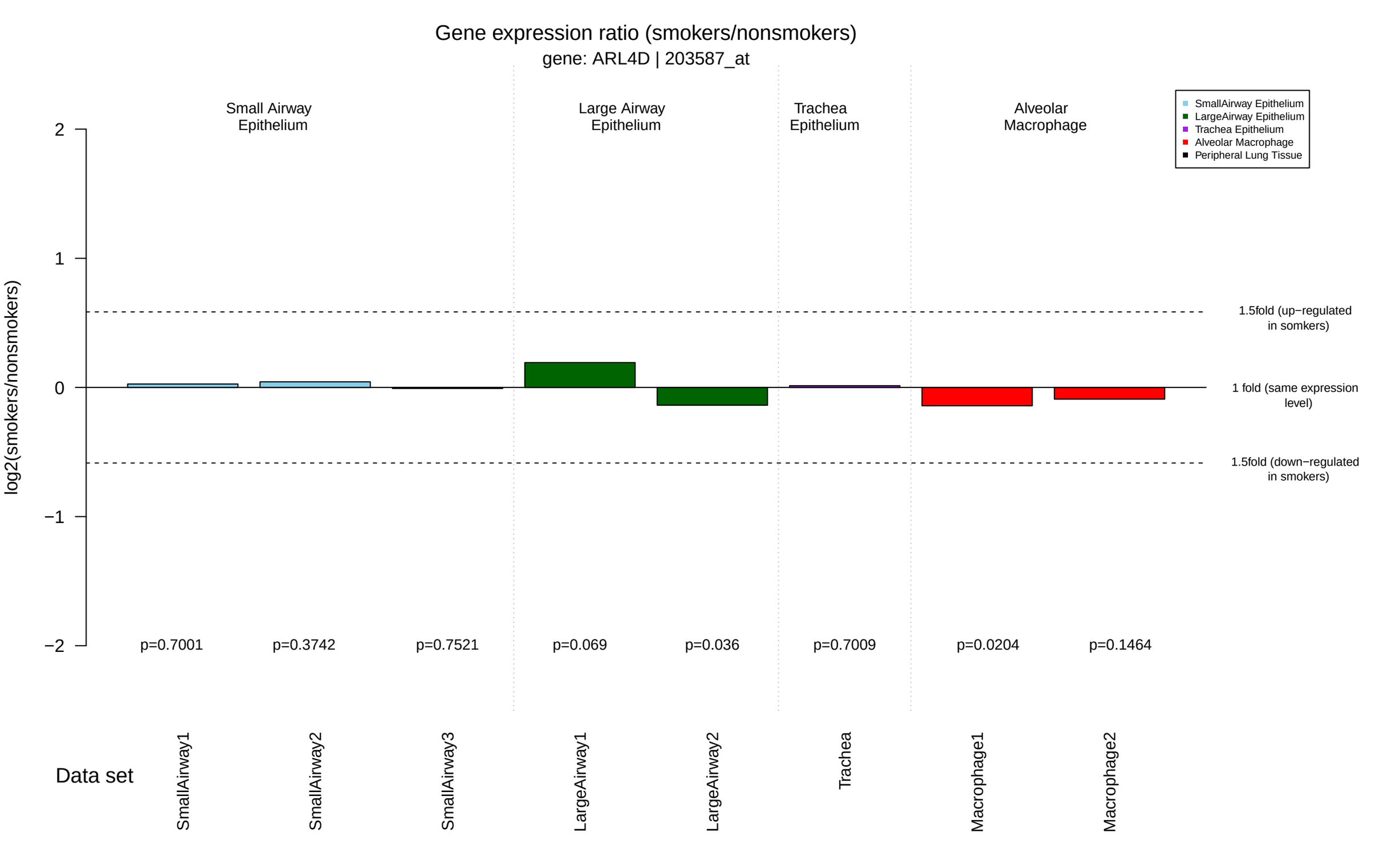
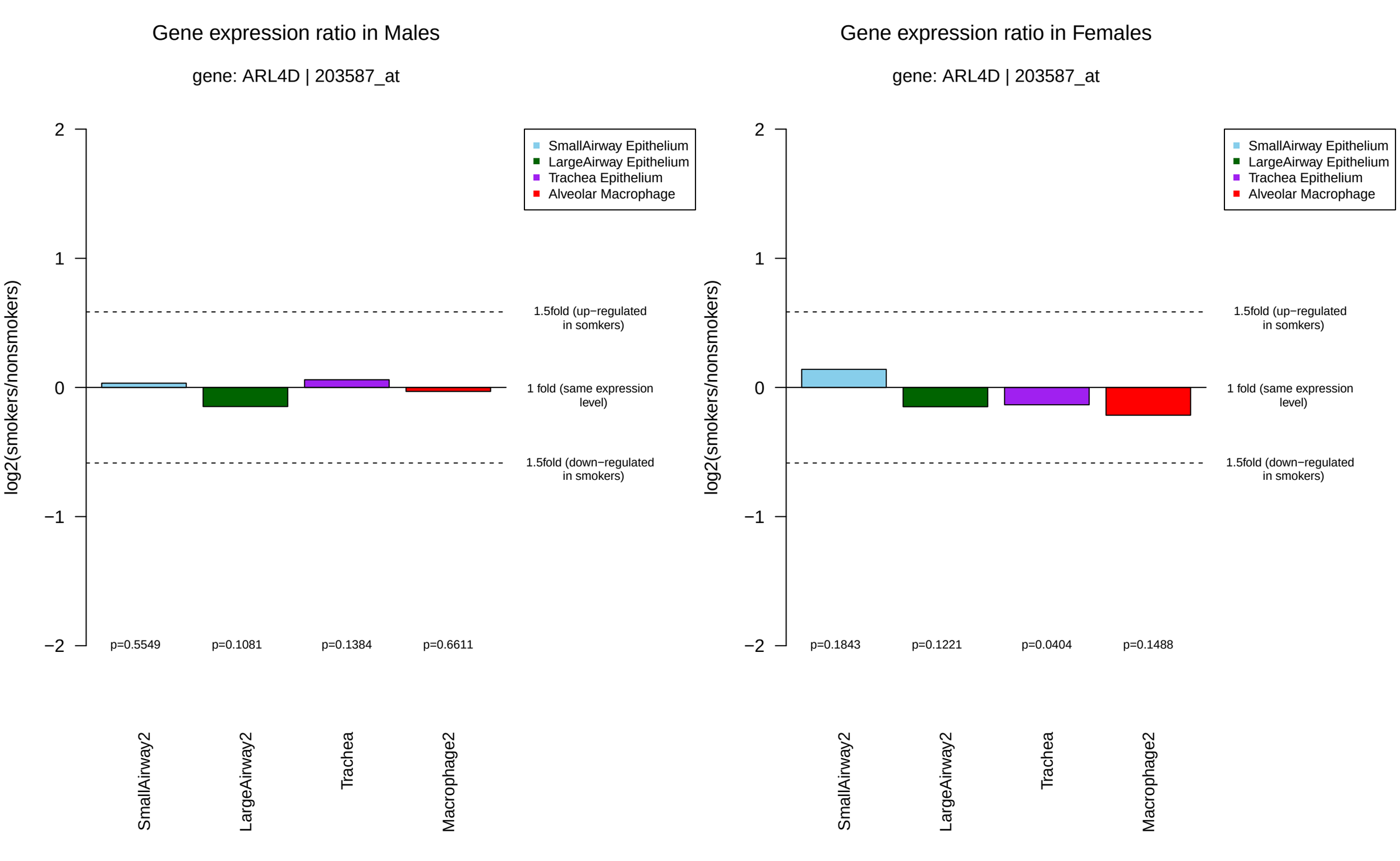
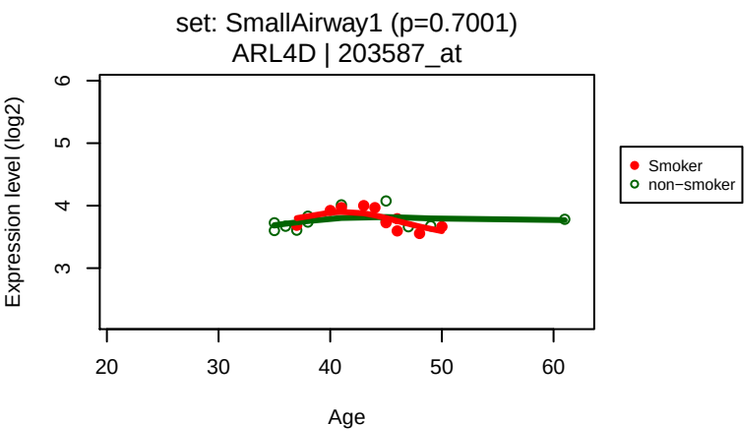
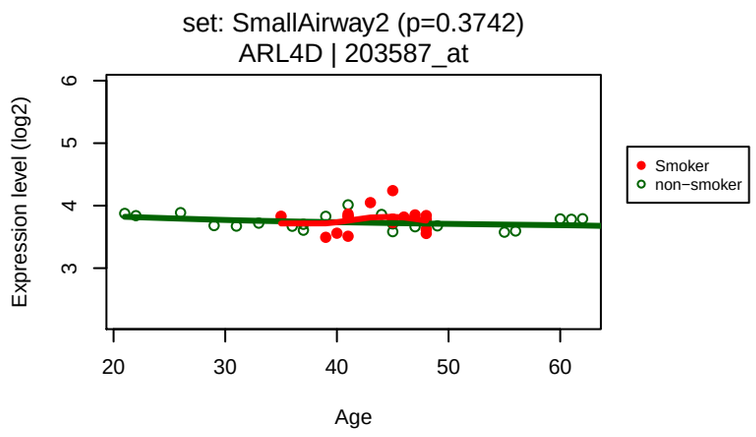
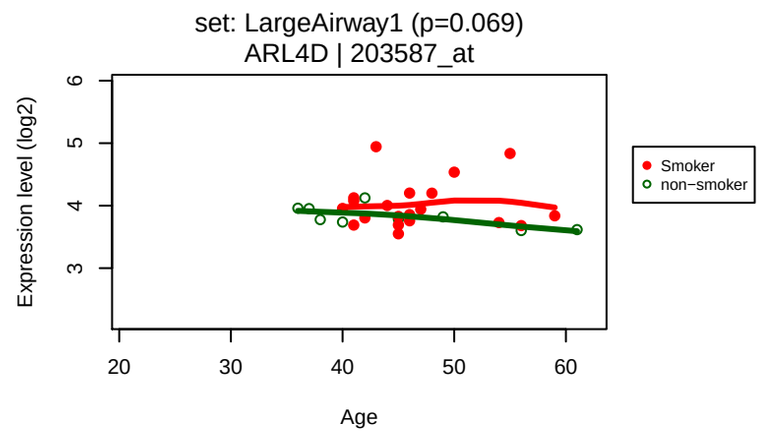

Gene expression of smokers and nonsmokers
gene: ARL4D | 203587_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7001
p=0.3742
p=0.7521
p=0.069
p=0.036
p=0.7009
p=0.0204
p=0.1464
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

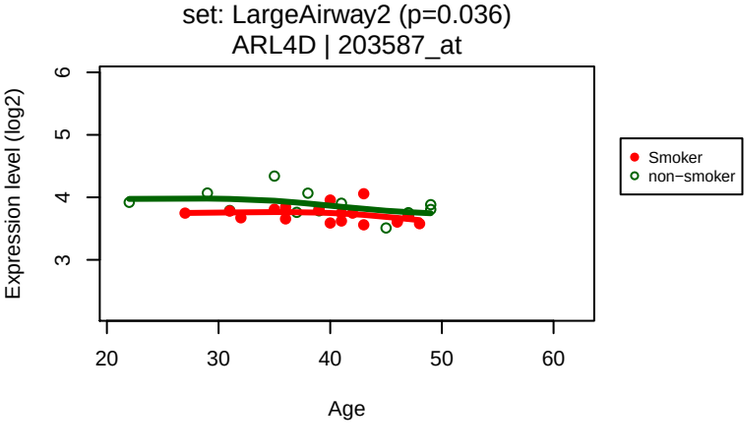

Gene expression ratio (smokers/nonsmokers)
gene: ARL4D | 203587_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7001
p=0.3742
p=0.7521
p=0.069
p=0.036
p=0.7009
p=0.0204
p=0.1464
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

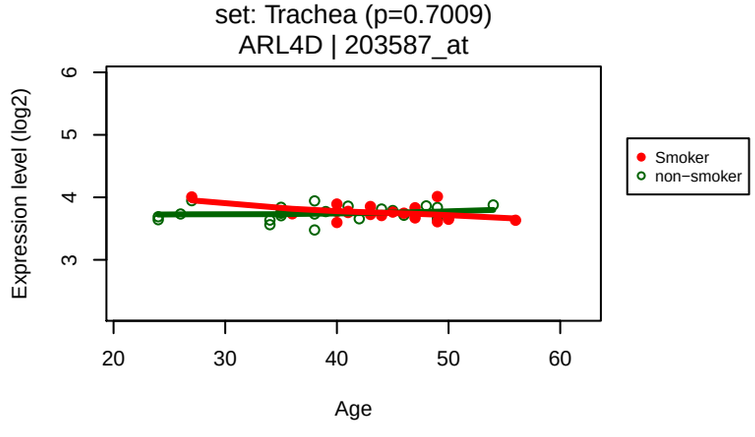

Gene expression ratio in Males
gene: ARL4D | 203587_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5549
p=0.1081
p=0.1384
p=0.6611
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL4D | 203587_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1843
p=0.1221
p=0.0404
p=0.1488
SmallAirway2
LargeAirway2
Trachea
Macrophage2

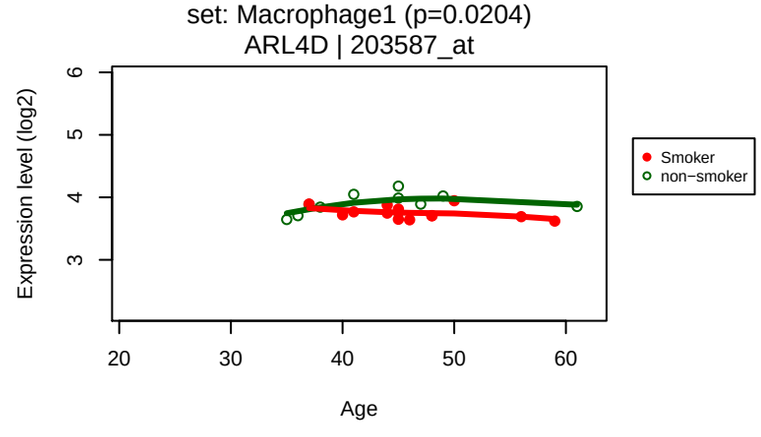

set: SmallAirway1 (p=0.7001)
ARL4D | 203587_at
Expression level (log2)
Age
Smoker
non−smoker

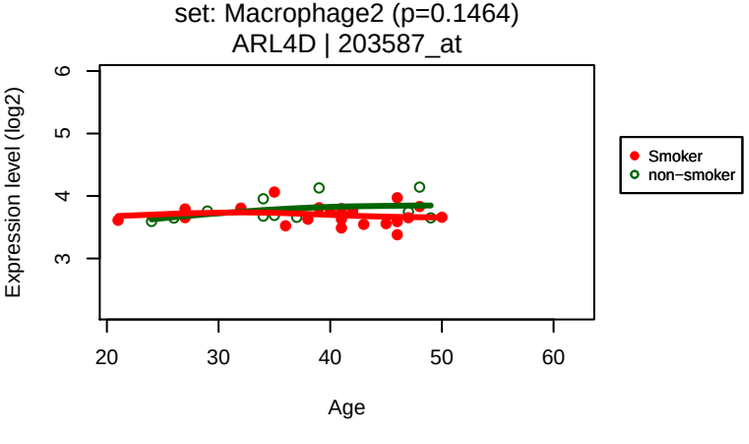

set: SmallAirway2 (p=0.3742)
ARL4D | 203587_at
Expression level (log2)
Age
Smoker
non−smoker

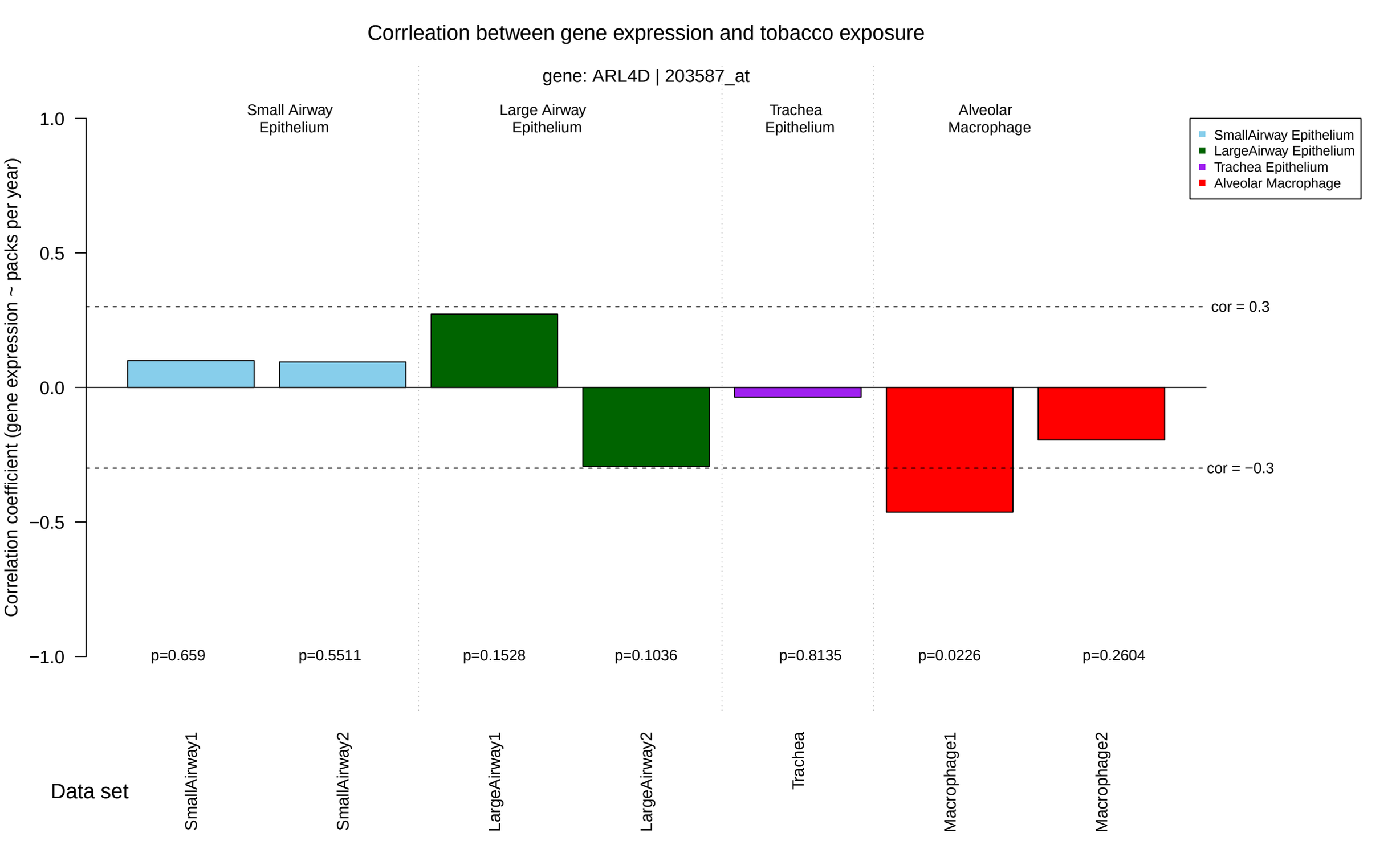

set: LargeAirway1 (p=0.069)
ARL4D | 203587_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.036)
ARL4D | 203587_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7009)
ARL4D | 203587_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0204)
ARL4D | 203587_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1464)
ARL4D | 203587_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL4D | 203587_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.659
p=0.5511
p=0.1528
p=0.1036
p=0.8135
p=0.0226
p=0.2604
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2